Final Project 2: Informed Search for Foraging Robot Swarms (100 pts)

Background: Central-Place Foraging Algorithm (CPFA)

The CPFA algorithm is inspired by the foraging behavior of ant colonies [1]. Its main purpose is to program a swarm of robots to autonomously forage for resources (e.g., minerals or survivors) in a large unknown environment and then deliver them to the central collection zone. Initially, a robot departs from the central collection zone and searches for resources randomly. If it finds resources, it will create a virtual pheromone waypoint based on the density of resources around that location. It will share this waypoint with the central server in the collection zone upon returning to the center. Once it arrives at the center, it drops off the collected resource and can access other pheromone information in the central server. It can decide where it should go based on the shared pheromone information in the central server, the last location where it found resources (called site fidelity), or search for resources randomly again. The behavior of a single robot is shown in Figure 1.

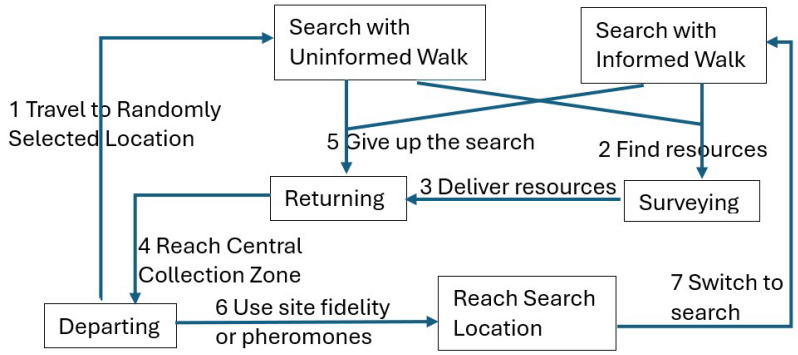


Figure 1: Individual robot's behavior in the CPFA

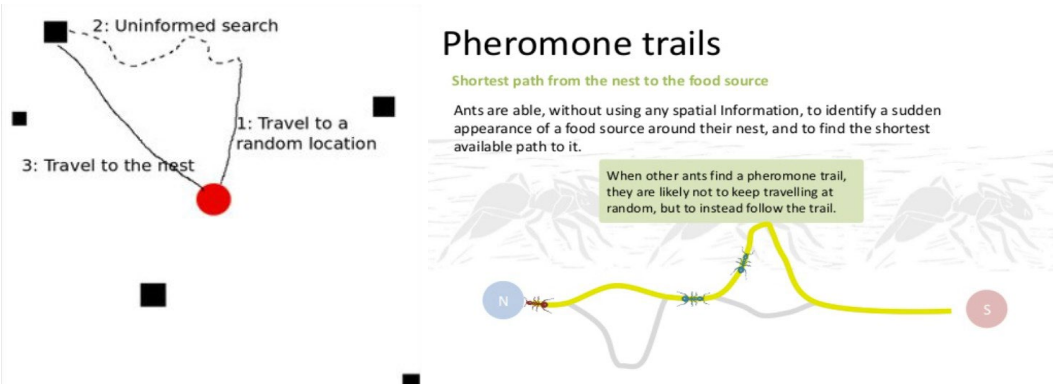


Figure 2: Sharing pheromone trails

Description:

In the current CPFA algorithm, robots share pheromone waypoints through the central server in the collection zone. Robots can find resources more efficiently by sharing the pheromone waypoints. Therefore, the locations of clustered resources are visited multiple times. Some other locations are never visited. If there are some resources in those unvisited locations, how can robots find them quickly?

If robots share their visited locations with the central server, the total number of locations is very large. For example, we have 200 robots, and every robot records the visited locations at a certain frequency (e.g., 5 seconds based on its sensing range). The maximum number of locations it can remember is 100. The total is 20,000 after one trip for all robots. If the average number of trips in a simulation is 20, the total is 400,000. The server must iterate this number of locations to find a new unvisited location for each robot when it returns to the center. The computation cost is very high.

Goals:

The goal is to find an efficient way to organize the number of locations. So, the server can find an unvisited location or region more efficiently. I shared one of my MS students' papers with you. He demonstrates a grid-based method for organizing locations. However, the grid map has a limitation when the search arena is large. The cell/region in the grid is large when the arena is large, which impacts efficiency.

You can think about whether you can cluster the locations more efficiently than grids. Clusters can be of different sizes based on the locations or points. You can also think about how to reduce the number of recorded locations, but still distinguish between visited and unvisited locations/regions. Your developed algorithm is still efficient.

The tasks are

1. You need to implement the algorithm to organize the locations efficiently. You can also think about other methods to improve the search strategy. This is the ultimate goal.
2. Evaluate the foraging performance in the three resource distributions (Random, Powerlaw, and Clustered). The following table shows the settings of the experiments.

Table 1: Experimental configuration

Size of the arena (m)	10 * 10
The radius of the collection zone (m)	0.25
Number of resources	256
Resource distributions	Random Powerlaw Clustered
Number of robots	24
Foraging time (minutes)	12
Runs in each distribution	50

3. Please compare your algorithm with the original CPFA algorithm and plot the data.

You can plot the results of the three distributions in a box plot. The data for each box plot is from 50 runs. You need to compare the performance to the original CPFA algorithm and report the improvement. Please show me the percentage improvement in the collected resources. I will compare your foraging scores to those of other teams. The team that has a higher score will get a higher grade.

CPFA Source code:

I am sharing my CPFA algorithm GitHub repository with you.
https://github.com/MARSLab-UTRGV/Collision_Free_CPFA/tree/main

Videos:

Multiple Place Foraging Algorithm:
https://www.youtube.com/watch?v=CoRMhYpC9gw